

Generation Interconnection Request System Impact Study Report

For Queue Project Q24-2464

Cataract Hydro 345 kV
September 17, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2464. The Interconnection Customer proposes a 188.3 MW solar plus storage facility interconnecting at the Cataract Hydro 345 kV substation in the Kentucky area. The request was received on October 16, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 457 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,668,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Delano-Kettle 138 KV Line
Contingency: McClellan Ar-Spatsum Lornex 138 KV
Loading: 127.8% (Rating: 227 MVA)
DFAX: 29.6%
Pre-project loading: 103.2%
- Facility: Sand Springs-Ojo 138 KV
Contingency: Northfield-Berry Street 345 KV
Loading: 107.2% (Rating: 528 MVA)
DFAX: 8.0%
Pre-project loading: 104.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kette-Delano 138 KV L	Rebuild line with higher-capacity conductor	\$12,348,000	45% (\$5,550,000)
Sand Springs-Ojo 138 KV	Reconductor line	\$2,878,000	4% (\$118,000)
	Total allocated to project		\$5,668,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.